Mutate a whole biological system

Complete Mutagenesis of the Genome and Proteome of ΦX174

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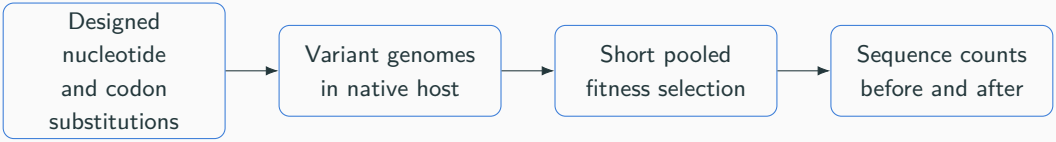
A genome we can read is not yet a system we understand

Question: can we explain and predict the fitness effect of changing every nucleotide and amino acid in a small, well-studied virus? ΦX174 has a 5,386-nt circular single-stranded DNA genome encoding 11 proteins, including overlapping open reading frames (ORFs). Its life cycle integrates replication, gene regulation, capsid assembly, host recognition, genome delivery and lysis. All mutations can therefore be measured against one system-level phenotype.

The central contribution
A near-complete single-substitution atlas connects molecular structures and variant predictors to viral fitness on a common experimental scale.

Introduction; Fig. 1a. "Complete" refers to single substitutions, not all combinations or environments.

One experimental design connects DNA changes to viral fitness



30 overlapping libraries tile the genome. Nucleotide substitutions and designed codon substitutions reach amino-acid changes that a single nucleotide cannot produce. Three independent biological replicates; 44,195 measured genotypes, over 99% of designs. Replicate Pearson correlations: 0.988–0.992.

Fig. 1b; Fig. S1a–d; Methods: variant library design, fitness selection and NGS processing.

What does the fitness number measure?

Input: assembled genome-library DNA. **Output:** intracellular variant genomes after approximately 80 minutes, spanning 2–3 wild-type generations. DiMSum estimates enrichment and uncertainty from sequencing counts. Libraries are calibrated using synonymous and nonsense modes:

$$F_{\mathrm{norm},i,l} = \frac{F_{i,l} - M_{\mathrm{nonsense},l}}{M_{\mathrm{synonymous},l} - M_{\mathrm{nonsense},l}}.$$

Here i is a variant and l a library. Shared protein anchors then align libraries; repeated measurements are combined using inverse-variance weights. **Interpretation:** a normalized pooled propagation phenotype, rather than a direct measurement of protein activity or binding energy.

Methods: fitness normalization, Eqs. 1–4; Fig. S1e. Normalization preserves gene-specific differences in nonsense effects.

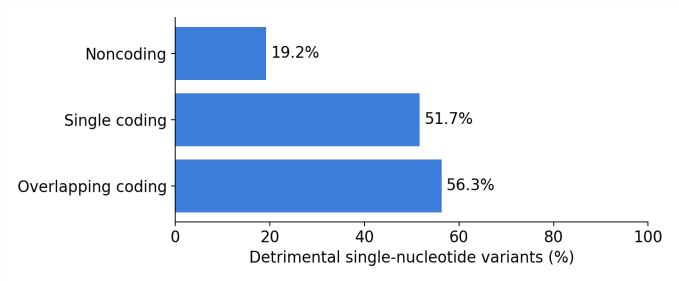
Translate statistical labels before reading the atlas

Detrimental: significantly lower fitness than the synonymous reference, Benjamini–Hochberg FDR < 0.05. **"Lethal"**: detrimental and a drop of at least 0.5 normalized fitness units. **Essential position**: at least one substitution at that position is classified as lethal.

Illustrative classification
Suppose the reference is 1.00 and a variant is 0.40. If significant, its 0.60-unit decrease meets the paper's "lethal" threshold. This label does not establish zero viable progeny, or that every substitution at that site is lethal.

Methods: statistical analysis and mutation-effect classification. Example values are pedagogical, not observations.

Half the single-nucleotide changes reduce fitness



8,229 / 16,098 nucleotide variants are detrimental (51.2%). 5,065 meet the "lethal" threshold. Overlapping coding regions are somewhat more sensitive; noncoding DNA is more tolerant overall.

Fig. 1c–d; bars redrawn from Results. Categories have different denominators: 636, 13,021 and 2,441 variants.

Regulatory overlap complicates a simple sequence explanation

Four noncoding regions occupy only 217 nt, but contain promoters, ribosome-binding sites (RBSs), terminators and an origin of replication. **81.1% of detrimental noncoding changes** fall in annotated regulatory elements. A promoter extension motif, TATG, has 9 detrimental changes among 10 measured substitutions. **A useful counterexample:** a terminator hairpin can show unequal sensitivity in its two stems when only one stem overlaps an RBS. The phenotype then reflects two molecular functions.

Interpretation
A DNA change can alter RNA structure, expression and replication simultaneously. A structural annotation is a hypothesis about its mechanism.

Fig. 1e–f; noncoding-genome Results; Fig. S1j. Promoter and overlap examples are reported observations.

A proteome-wide map still needs careful denominators

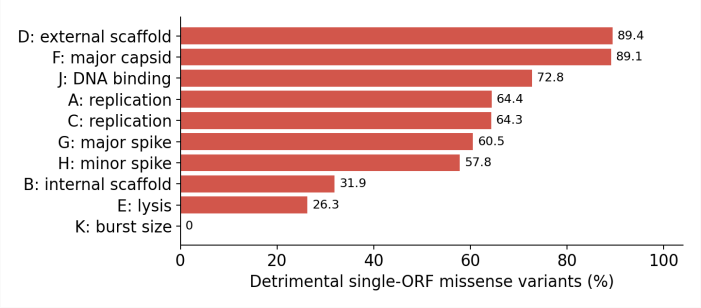
39,719 / 39,920 possible single-amino-acid mutations were measured (99.5%): 29,979 single-ORF and 9,740 multi-ORF mutations. **Most mechanistic analyses exclude multi-ORF changes.** Within the remaining 28,380 missense substitutions:

Category	Count	Fraction
Detrimental	19,908	70.1%
"Lethal" subset	12,658	44.6%
Fitness-increasing	403	1.4%

Control behavior: 92% of synonymous variants are neutral; 99% of nonsense variants are classified as lethal, with protein K an exception.

Fig. 2a–b; Results: proteome landscape and synonymous variants. Genotype counts and amino-acid counts are not interchangeable.

Mutation tolerance is highly protein dependent



Scaffold D and capsid F tolerate few substitutions. K shows no detectable detrimental mutations in this assay. E and K are cautionary: nearly 90% of their changes affect other ORFs and are excluded from this comparison.

Fig. 3a; values redrawn from Results. These are mutation-level percentages, not the fraction of essential residues.

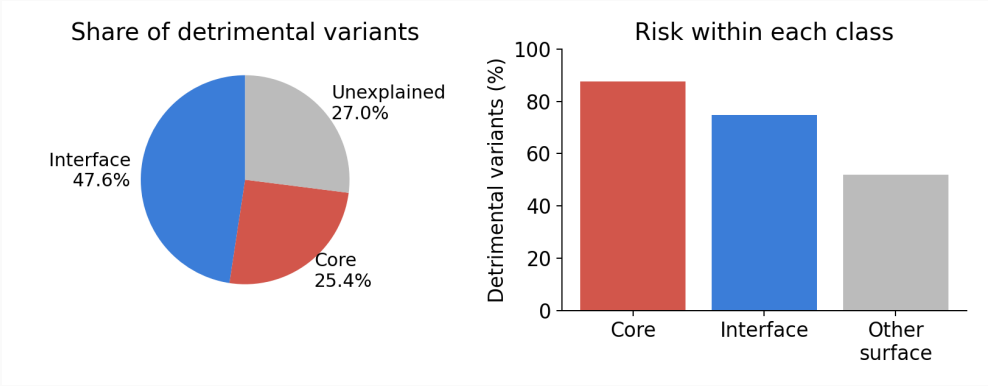
The mechanistic test: where are damaging changes located?

Protein core: relative solvent accessibility (RSA) $< 25\%$ in a monomer structure. **Interface**: surface residue with a heavy atom within 5 Å of another protein or DNA in a complex. **Other surface**: exposed, without an annotated interaction partner.

Experimental capsid, tunnel and host-complex structures supply known contacts. AlphaFold 3 adds models of replication-associated proteins and complexes. Mechanistic coverage rises from 54% to 73% of detrimental substitutions after adding predicted structures. Location is evidence of a plausible mechanism; it is not a direct perturbation assay of that mechanism.

Fig. 4a–b; Methods: protein structure analyses; Table S1. Classes separate core from surface interfaces.

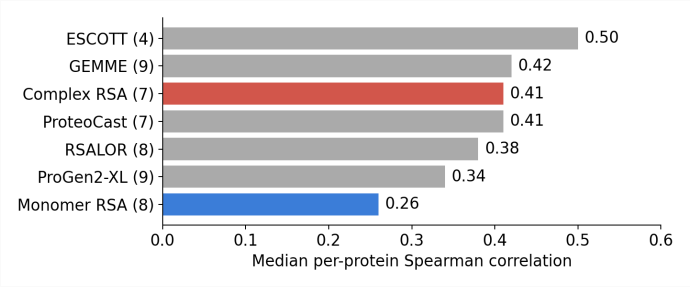
Interfaces explain more cases; cores have higher per-change risk



Interface variants are more often damaging than other surface variants (74.8% vs. 52.0%; odds ratio 2.73). Core variants have still higher risk (87.5%). The larger interface contribution reflects **more residues participating in interactions**, rather than greater vulnerability of each individual substitution.

Fig. 4b–e; redrawn. Denominator for the left panel: 19,908 detrimental single-ORF missense changes.

An interaction-aware structural baseline is hard to beat



Rank correlation is evaluated separately for each protein. Complex RSA captures burial by interaction partners; monomer RSA misses it. Numbers in parentheses indicate proteins evaluated. Methods have different coverage, so median rankings are not strictly matched comparisons.

Fig. 4h; Fig. S4f; Methods: VEP evaluation. Selected methods shown, not the complete benchmark.

The unexplained quarter is biologically informative

27% of detrimental substitutions remain unexplained by current core and interface annotations: 5,372 changes. 2,570 fall in regions lacking adequate structural information.

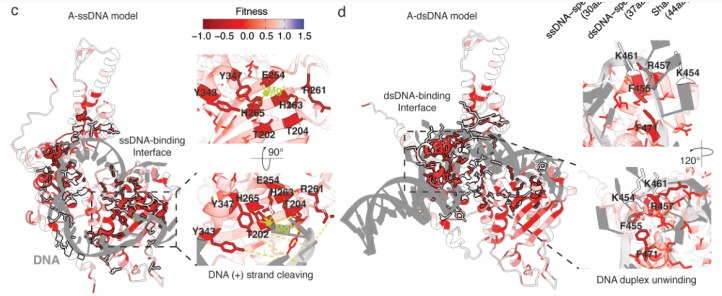
Protein H accounts for 2,370 of these: the known tunnel structure covers only part of its sequence. The other 2,802 occur on annotated non-interface surfaces. Possibilities include missing partners, regulatory functions or conformational requirements.

An informative exception

In DNA-binding protein J, unexplained damaging surface substitutions are strongly enriched at glycine. Backbone flexibility may matter even where burial and known contacts do not explain sensitivity.

Fig. 4b; Fig. S4g–h; Table S1; Results: unknown molecular mechanisms. Proposed explanations remain hypotheses.

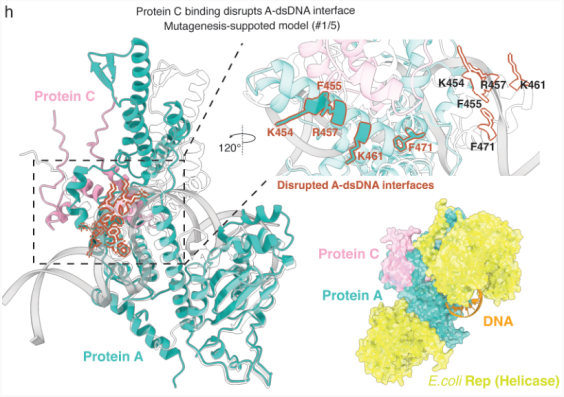
Replication protein A: distinct DNA-binding modes



AF3 predicts distinct ssDNA and dsDNA contacts. The ssDNA model includes known catalytic tyrosines; the dsDNA model suggests aromatic residues that wedge apart the duplex. Damaging changes are enriched in ssDNA-specific, dsDNA-specific and shared interfaces (odds ratios 3.55, 1.76 and 1.92). This supports functional interfaces, without directly proving the proposed unwinding mechanism.

Fig. 5c–e, cropped c–d (Wei et al., CC BY 4.0); Fig. S5a–c. DNA grey; protein colored by median fitness.

Protein C regulation: a hypothesis supported by mutagenesis



150 AF3 models span A–C, A–C–DNA and assemblies including Rep helicase. Five models show damaging-interface enrichment on both A and C; all contain Rep. C lies near A's proposed dsDNA interface, suggesting inhibition of DNA engagement while preserving the catalytic site. All A–C interface confidence scores are low. The illustrated supported model has ipTM 0.27.

Fig. 5h (crop, CC BY 4.0); Fig. S5e–k. Structural selection using the same mutational data motivates independent tests.

What is established, and what remains to test?

Established in this assay: near-complete substitution coverage, reproducible fitness effects, strong protein-specific constraints and enrichment of damaging changes at interfaces. **Limits:**

one host and selection regime; pooled short-term propagation; single substitutions; exclusion of many overlapping-ORF changes; incomplete and sometimes low-confidence structural models. **Our interpretation:** the strongest result is a system-level experimental benchmark.

Attributing an individual change to disrupted binding or folding still needs targeted validation.

Discussion
Would matched-coverage predictor comparisons alter the ranking? Can binding, replication and rescue experiments distinguish the proposed A–C mechanism from other explanations for the same fitness pattern?

Discussion; Methods; Figs. 4–5 and S5. Independent mechanism tests are journal-club proposals.

Sources and reusable takeaways

Wei H, Li X, Lehner B. Complete Mutagenesis of the Genome and Proteome of ΦX174. bioRxiv (2026), version 1, posted 26 July 2026.
`https://doi.org/10.64898/2026.07.25.740675`

These notes use the complete 49-page preprint, including Methods, supplementary Tables 1–2 and supplementary Figs. S1–S5. **Takeaway:** complete single-substitution maps reveal what structural models explain, where predictors fail and which mechanistic experiments are most informative.

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